

SEDAR 23: FWC Goliath Grouper

Data Workshop

LIFE HISTORY WORKING GROUP

Stock Definition:

- D. Jones has new MARFIN funding to use otolith microchemistry (laser ablation) to determine if there are distinct subpopulations based on geographic differences in chemical signatures. Juvenile habitat would be represented at the origin of otolith, adult habitat at the margins (SA and/or Gulf) ***goliath grouper were not originally considered in this MARFIN proposal, but could easily be added with availability of otoliths and moderate time resources.*
- Koenig referenced the availability of goliath grouper eggs from the SA and GOM which could be used for genetic population structure analysis. Eggs will be sampled for Dr Matthew Craig (U Puerto Rico) who has done the most extensive work on goliath grouper population genetics (Craig et al. 2009)
- Description of larval stages of goliath grouper is part of an ongoing MARFIN project by Koenig and Coleman.
- Limited recent drifter studies along the US South Atlantic coast have shown the potential for wide distribution patterns along the coast from Cape Hatteras to the Florida Keys (Leshner and Sedberry, SEDAR 10-DW-06). With location and timing of spawning now known, it would be a good opportunity to initiate additional drifter studies in the SA and GOM.
- Ongoing research (Koenig and Coleman) will verify known SPAGS and suspected SPAGS. It will also determine the size structure of spawning fish, their residency time on the SPAGS, and size-related fecundity. With more known SPAGS, there is the potential to assess the abundance of reproductive adults based on numbers present at SPAGS and knowing the geographical range of the participating spawners.

Age and Growth:

- A directed effort to collect hard parts from large, old fish to validate these methods for old individuals.
- More detailed information on maximum age and size is needed. There are no new data available for maximum age or maximum size since Bullock et al. 1992. There is reason to suspect that maximum age is a low estimate due to the small number of large, old fish sampled. Additionally, there is concern over whether or not the asymptote is fully represented due to the low number of samples represented at the oldest ages (Fig.1). However, this maximum age does fall within the values observed for other epinephelines [i.e., *E. fuscoguttatus* (42 y for females and 40y for males; Pears, 2006), *E. morio* (29 y; Lombardi-Carlson et al., 2006), *E.*, *H. nigrus* (41 y; Manooch, 1987), *E. striatus* (29 y; Sadovy and Ecklund 1999)]. However, the best species for

comparison (due to similar size, tropical/subtropical distribution and ecological role) are the Indo-Pacific *E. lanceolatus* and *E. tukula*; data on maximum size, age and growth rate are still being sought at the time of writing the present report.

- As suggested during the last SEDAR (SEDAR6, 2004): “The panel recommended continued work on ageing. Ages should be standardized to a calendar year, so that information on a year class is treated consistently throughout the year.”

Reproduction

- Ongoing research (Koenig and Coleman, MARFIN) will evaluate fecundity, sexual pattern, SPAG distribution, size structure and sex ratio within SPAGS, and mating system using non-lethal methods.

Habitat and Movement:

- We need spatially-explicit models. Due to microhabitat preferences and site attachment in both juvenile and adult goliath groupers, density values (as number of individuals per unit area or length of coastline) should be used with caution in population estimates and modeling; it is essential to contrast densities in high quality habitats versus low quality habitats, and not use a single density value which could result in over-estimates of total population levels. Future modeling efforts should also account for the known (or unknown) statewide spatial distribution of both juveniles and adults.
- We need a state-wide evaluation of habitat quality integrating habitat structure and water quality. Including this knowledge in our goliath grouper assessments will allow us to expand population models into ecosystem-based management.
- What is the extent of high quality mangrove habitat, and where is it located in Florida? There is a need for a state-wide assessment of mangroves as fish habitat, to evaluate potential high quality sites that are the nurseries, not only for juvenile goliath grouper but also for juveniles of a diverse group of other fish and invertebrate species.
- When evaluating high quality habitat (both in mangroves and reefs), in addition to evaluating the structural characteristics, what is the water quality of each habitat? There is a need to quantify, state-wide in real time and 24/7 the water quality (salinity, temperature, dissolved oxygen) of mangroves, and coastal reefs. This research question applies not only to goliath grouper but also to all estuarine and coastal species that use mangroves and reefs (coral reefs, reef ledges) during their life history.
- What are the biological corridors used during the ontogenetic migrations (from juvenile mangrove habitat to reef adult habitat) and the spawning migrations (from resident habitat to spawning aggregation sites)? We don't know if goliath grouper use a specific path or network (=biological corridor) during their two major migratory events (ontogenetic and reproductive).

- What are the maximum distances that can be covered by juveniles in ontogenetic migrations towards the adult habitat, and by adults in their spawning migrations? These data are needed to understand the ontogenetic and spawning connectivity within the goliath grouper population.

COMMERCIAL AND RECREATIONAL STATISTICS WORKING GROUP

The prohibition on any harvest of goliath grouper precludes any fishery dependent research other than that conducted by on-board observers or recorded in fishermen's logbooks. Continued collection of size, frequency in the catches by gear, and observed release condition is important for obtaining release mortality estimates and possibly an estimate of numbers caught by gear, fishing area, and depth. It is expected that as the abundance of this species increases, so too will the frequency of encounter with fishing gears. Brusher and Schull's (2009) study that goliath grouper have a reasonably good chance of surviving the encounter with fishing gear at least in shallower waters. Capture-recapture studies could be designed to examine the effects of releases from the recreational fishery. With the apparent increase in numbers of goliath grouper reported by anglers, it is inevitable that more encounters with fishing gear will occur and this seems to be borne out by reports from angler surveys such as the ENP Angler Creel Survey and the MRFSS. Surveys of spawning aggregations are needed to extend the usefulness of Don DeMaria's earlier surveys and to monitor population trends of adults.

INDICES OF ABUNDANCE WORKING GROUP

No research recommendations were provided by the Working Group.

Assessment Workshop

Recommendations on future research and data collection were provided at the DW. An additional recommendation for a "research fishery" was briefly discussed at the Assessment Workshop (AW), but was never formalized.

Review Workshop

Although results were unsatisfactory for this stock assessment, they did serve to clarify additional research necessary for future assessment efforts. The next benchmark assessment cannot be successfully completed without data from the research recommended by the Data, Assessment, and Review Panels.

Stock Definition:

- Goliath grouper should be genetically sampled from as many areas in the South Atlantic and Gulf of Mexico as possible to allow for a more thorough examination of the current single stock definition.
- Examination of spawning aggregations over the entire distribution range should include seasonality, sex ratios, and individual fidelity.

Long-term monitoring:

- Basic reproductive data are lacking throughout the species distribution, including: size and age at maturity for each sex, sexual sequence with size and age for each sex, and fecundity.
- As described in the above research recommendations by the Life History Working Group, research on age structure, and locations of suitable juvenile and adult habitat, discard and discard mortality rates should be accomplished throughout the species distribution

Economic impact:

- Because of the relatively small size of a potentially reopened consumptive fishery for goliath grouper, a socio-economic evaluation of the relative benefits of consumptive versus non-consumptive uses would be beneficial. There may be greater long-term economic benefit to development of sustainable non-consumptive eco-tourism venues than would be possible from a consumptive fishery.